Fetal period
3rd month-Birth

By

Prof. Dr. Ehab Elzawawy

TW Sadler. Third month to birth: In Langman's medical embryology, 15th edition,2024, Chapter 8: P108-128.

Learning objectives

- The amniotic cavity
- Growth in length and weight.
- Changes in fetal measurements.
- Development in the 3rd and 4th months.
- Development in the 5th and 6th months.
- Development in the 7th month.
- Development in the last 2 months.
- The newborn at time of birth.
- How to determine length of pregnancy.
- Define low birth weight, small for gestational age and intrauterine growth restriction.

The amniotic cavity

- At the end of the 2nd month, there are 2 cavities: **the amniotic cavity** and **the chorionic cavity**.
- At the end of the third month, the amniotic cavity has expanded so that it obliterates the chorionic cavity.

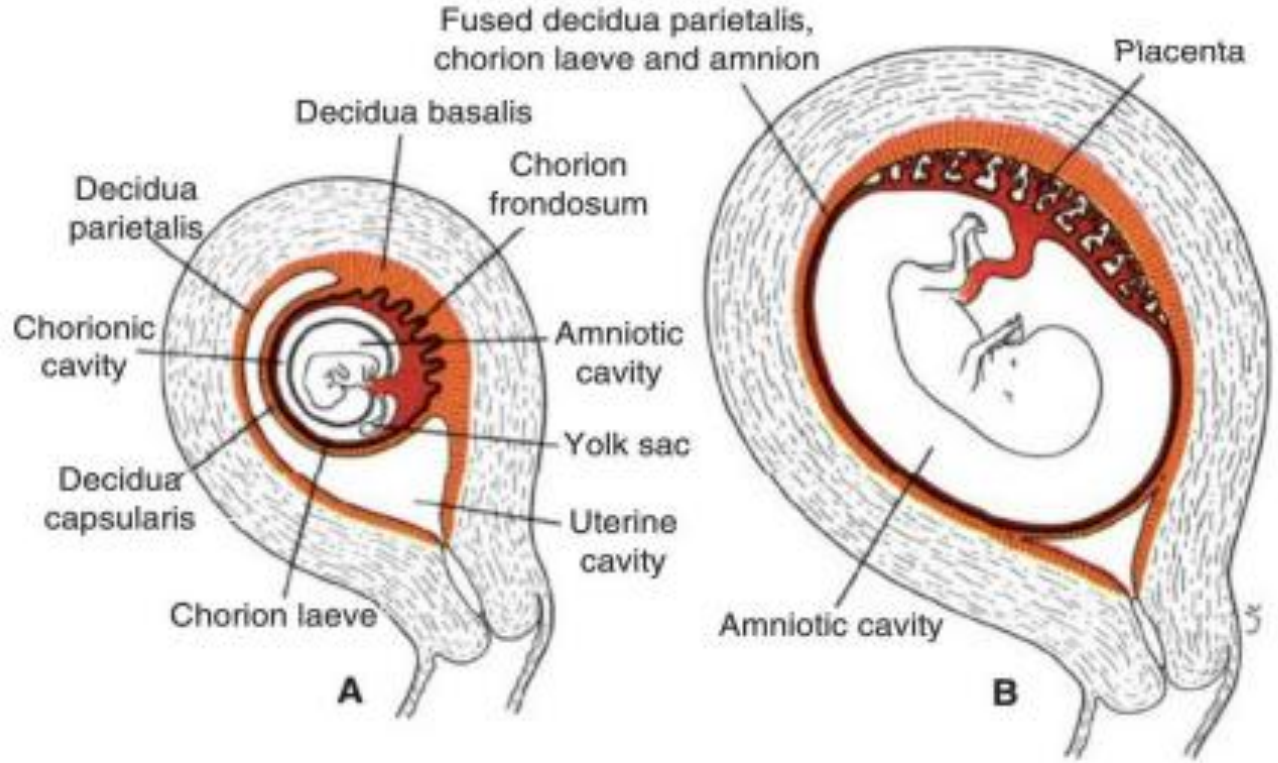


FIGURE 8.10 Relation of fetal membranes to wall of the uterus. **A.** End of the second month. Note the yolk sac in the chorionic cavity between the amnion and chorion. At the abembryonic pole, villi have disappeared [chorion laeve]. **B.** End of the third month. The amnion and chorion have fused, and the uterine cavity is obliterated by fusion of the chorion laeve and the decidua parietalis.

The amniotic cavity

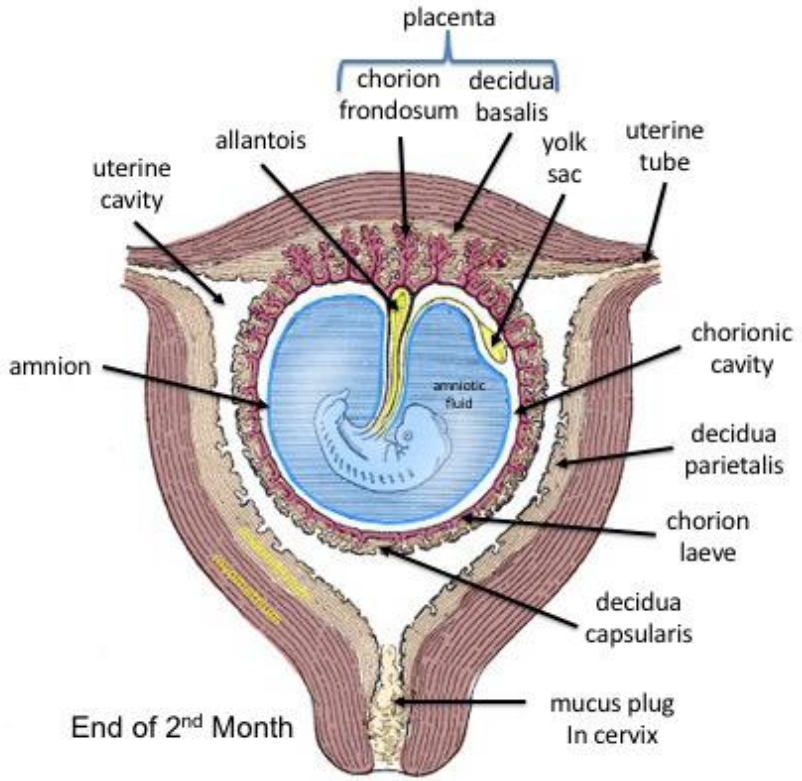
- The amniotic cavity is filled with a clear, watery **amniotic fluid** that is produced in part by amniotic cells and maternal blood.
- The amount o f fluid increases from 30 mL at 10 weeks to 450 mL at 20 weeks to 800 to 1,000 mL at 37 weeks.
- **Functions** of amniotic fluid fluid:

(1) Protective cushion.

(2) Prevents adherence of the embryo to the amniotic membrane.

(3) Allows for fetal movements.

(4) During childbirth, the amniotic membrane helps to dilate the cervical canal.



The amniotic cavity

- From the beginning of the fifth month:

1- The fetus swallows its own amniotic fluid, and it is estimated that it drinks about 400 mL a day at full term, about half of the total amount.

2- Fetal urine is added daily to the amniotic fluid, but this urine is mostly water because the placenta is functioning as an exchange for metabolic wastes.

The volume of amniotic fluid is replaced every 3 hours.



Polyhydramnios

- Polyhydramnios is the term used to describe an excess of amniotic fluid (1,500 to 2,000 mL).
- Primary causes of polyhydramnios include **idiopathic causes (35%), maternal diabetes (25%), central nervous system malformations (e.g., anencephaly) and gastrointestinal defects (esophageal atresia)** that prevent the infant from swallowing the fluid.
- Prematura rupture of the membranes (PROM) refers to rupture o f the membranes before uterine contractions begin and occurs in 10% of pregnancies.



Oligohydramnios

- Oligohydramnios refers to a decreased amount (<400 ml).
- Oligohydramnios is a rare occurrence that may result from **renal agenesis.**
- The lack of fluid in the amniotic cavity may constrict the fetus.
- The fetus is unable to " breathe" into his lungs resulting in **lung hypoplasia.**
- **Amniotic bands.**

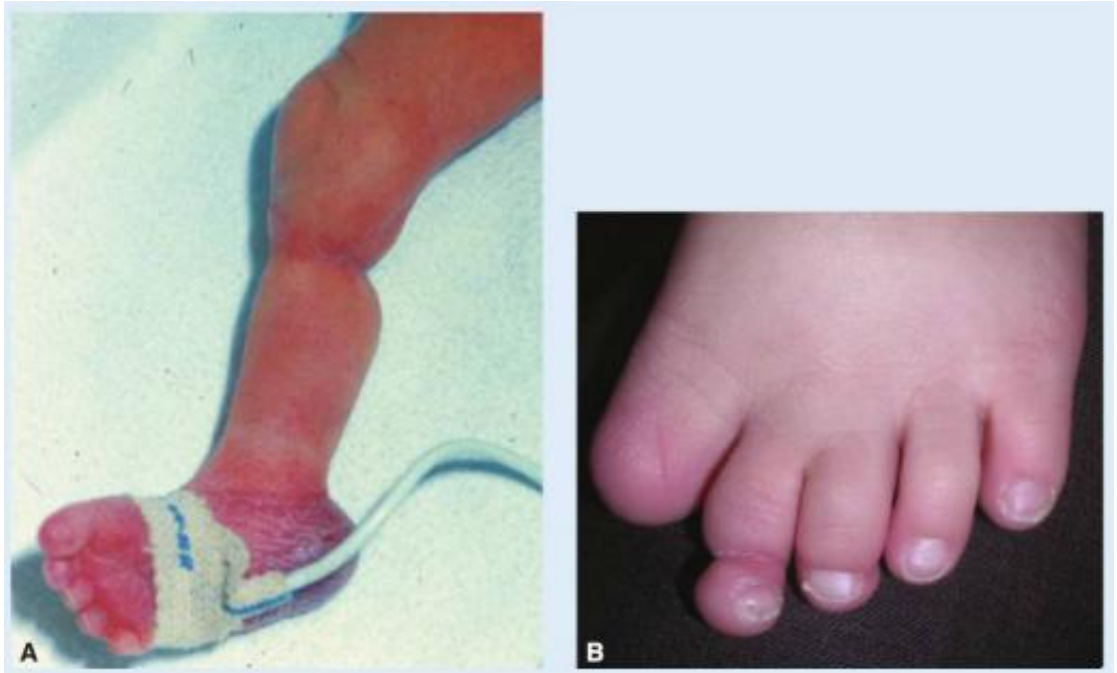


FIGURE 8.17 Limb abnormalities caused by amniotic bands. **A.** Limb constriction ring. **B.** Digit amputation [big toe] and ring constriction [second toe].



Main events in the fetal period

- Maturation of organs.
- Rapid growth of the body.
- Growth in length is particularly striking during the third, fourth, and fifth months.
- Increase in weight is most striking during the last 2 months as 50% of the full-term weight is added.

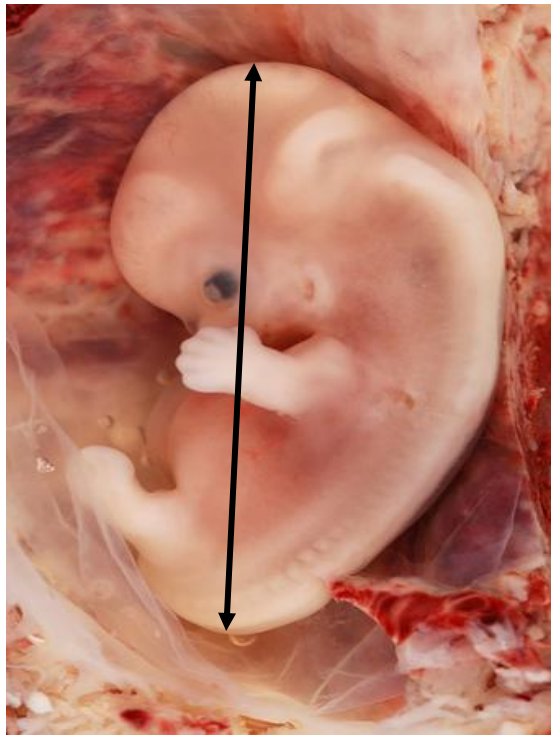
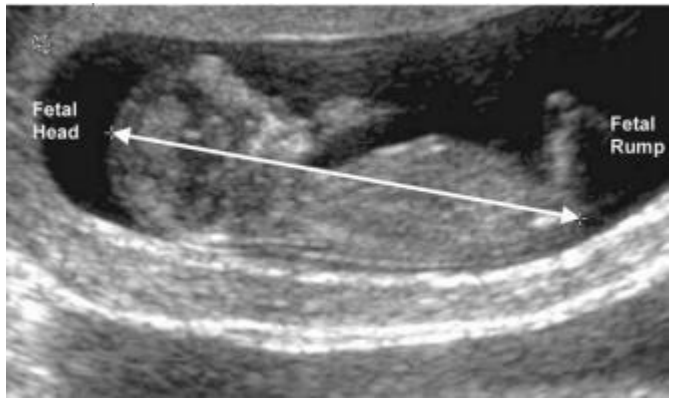


TABLE 8.1 Growth in Length and Weight during the Fetal Period

Age (wk)	CRL (cm)	Weight (g)
9-12	5-8	10-45
13-16	9-14	60-200
17-20	15-19	250-450
21-24	20-23	500-820
25-28	24-27	900-1,300
29-32	28-30	1,400-2,100
33-36	31-34	2,200-2,900
37-38	35-36	3,000-3,400

CRL, crown-rump length. (Sitting height)



Growth of the head and body

- At the beginning of the third month, the head constitutes approximately half of the CRL.
- At the beginning of the fifth month, the head is about one third of the CHL.
- At birth, the head is approximately one quarter of the CHL.
- Hence, over time, growth of the body accelerates but that of the head slows down.

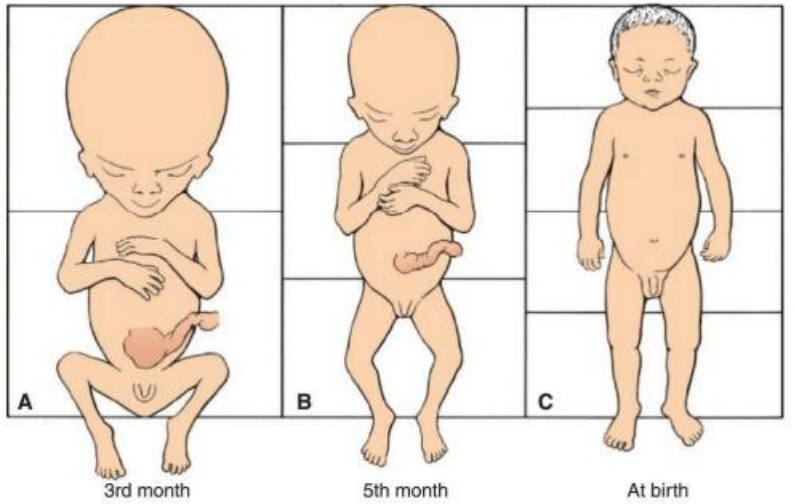


FIGURE 8.2 Size of the head in relation to the rest of the body at various stages of development.



Figure 3: Crown heel length.

During the 3rd & 4th month

- The face becomes more human like.
- The limbs increase in length.
- By the 12th week, primary ossification centers are present in the long bones.
- By the 12th week, external genitalia develop, and the sex of the fetus can be determined by ultrasound.



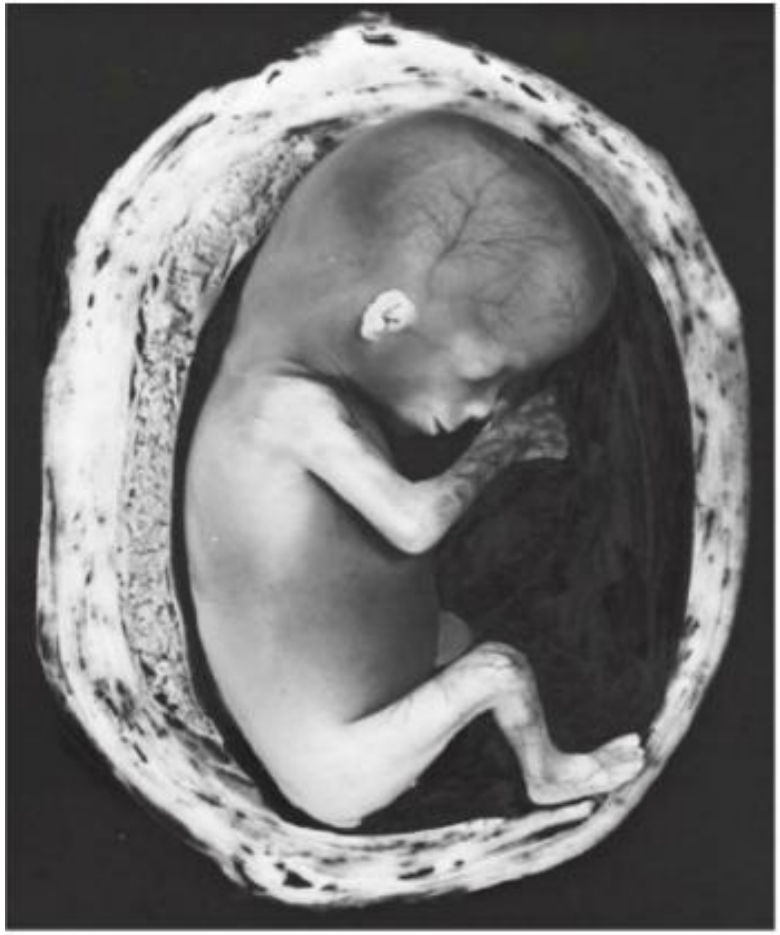


FIGURE 8.4 A 12-week fetus in utero. Note the extremely thin skin and underlying blood vessels. The face has all of the human characteristics, but the ears are still primitive. Movements begin at this time but are usually not felt by the mother.

During the 5th month

- The fetus is covered with fine hair, called lanugo hair.
- The eyebrows and head hair are visible.
- The movements of the fetus can be felt by the mother (quickening).



FIGURE 8.5 An 18-week fetus connected to the placenta by its umbilical cord. The skin of the fetus is thin because of lack of subcutaneous fat. Note the placenta with its cotyledons and the amnion.

During the 6th month

- The skin of the fetus is reddish and thin due to lack of subcutaneous fat.
- A fetus born early in the sixth month has great difficulty surviving.
- Although several organ systems are able to function, the respiratory system and the central nervous system have not developed sufficiently, and coordination between the two systems is not yet well established.



TABLE 8.2 Developmental Horizons During Fetal Life Event

	Age (wk)
Taste buds appear	7
Swallowing	10
Respiratory movements	14–16
Sucking movements	24
Some sounds can be heard	24–26
Eyes sensitive to light[a]	28

During the 7th month

- By 26-28 weeks, the fetus has a CRL of about 25 cm and weighs approximately 1,100 g.
- If born at this time, the infant has a 90% chance of surviving.

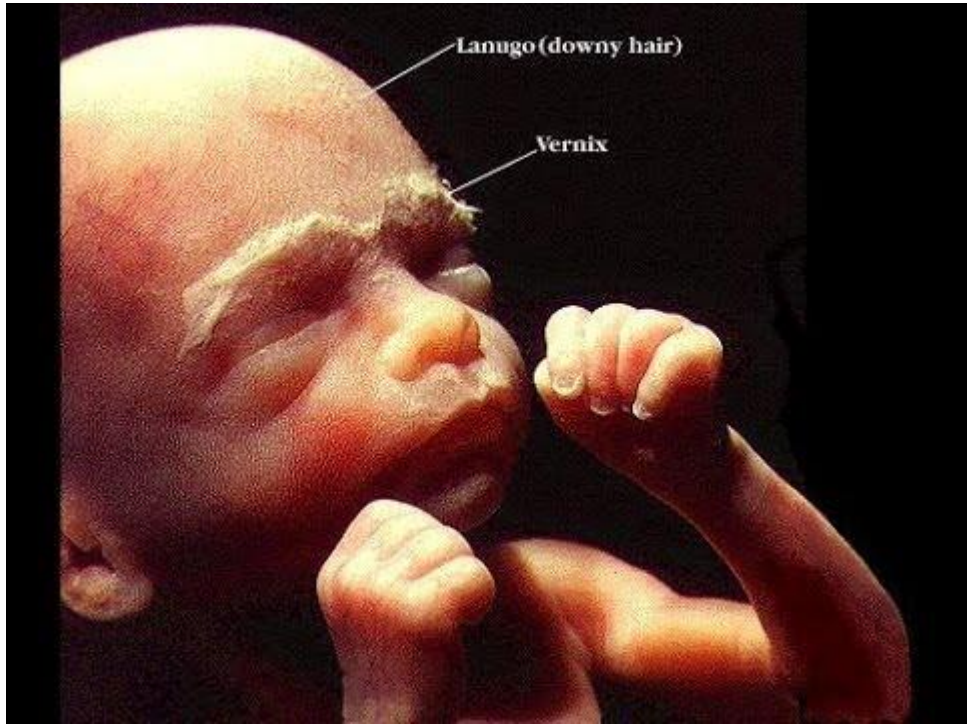


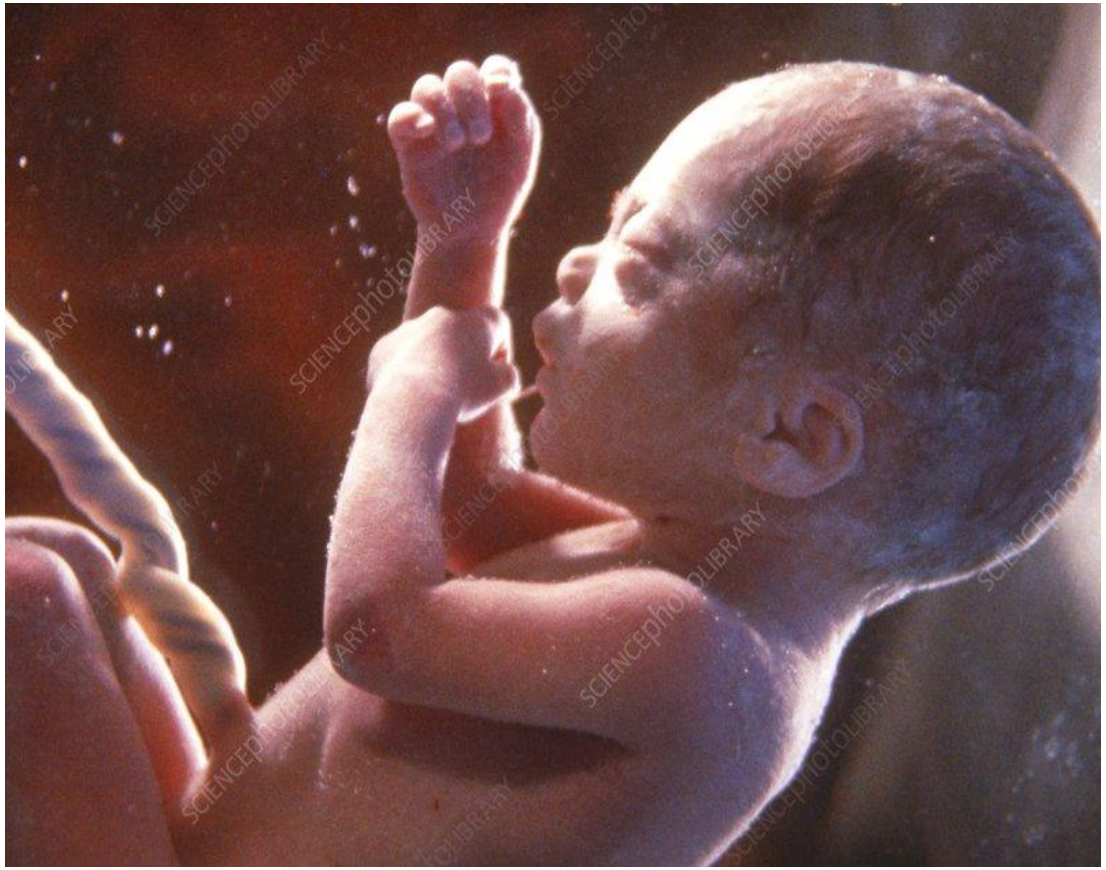


NICU

During the last 2 months

- The fetus obtains well-rounded contours as the result o f deposition o f subcutaneous fat.
- At full term, the skin is covered by a whitish, fatty substance (vernix caseosa) composed of secretions from sebaceous glands.





Baby has well-rounded contours as a result of deposition of subcutaneous fat. Note the twisting of the umbilical cord. This fetus would be able to survive.

At the time of birth

- The average weight is 2,500 to 4,000g (Appropriate for gestational age), mostly between 3,000 to 3,400g.
- CRL is about 36 cm, and its CHL is about 50 cm.
- The skull has the largest circumference of all parts of the body (Head circumference: 32-38), an important fact about passage through the birth canal.
- The testes should be in the scrotum.



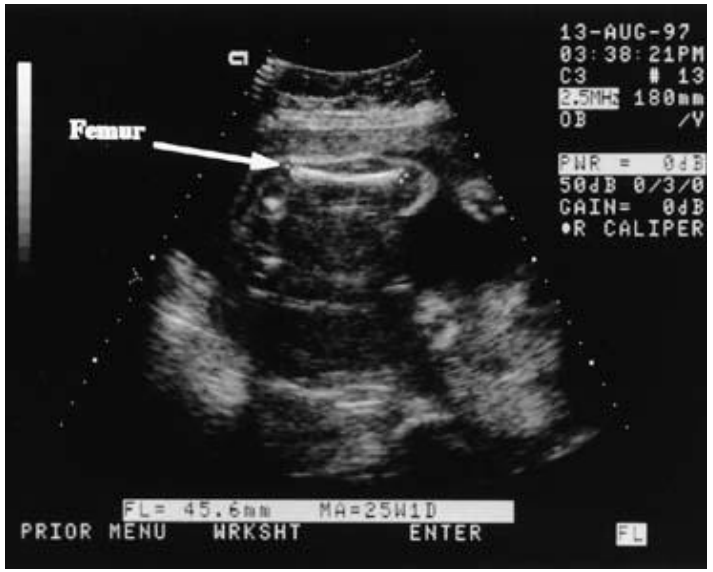
Length o f pregnancy (Time of birth)

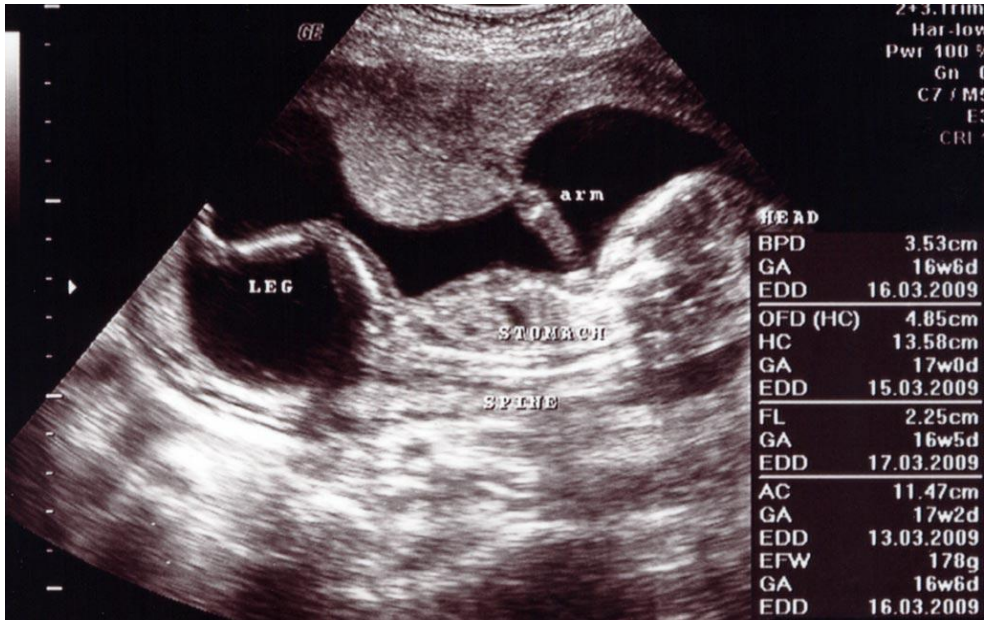
- 280 days, or 40 weeks after the onset of the last normal menstrual period (LNMP).
- More accurately, 266 days or 38 weeks after fertilization.
- Full-term: 37-42 weeks.
- Preterm: less than 37 weeks.
- Late preterm: 34 weeks to less than 37 weeks.
- Post-term: after 42 weeks.

Role of Ultrasound

- A valuable tool for the determination of the length of pregnancy.
- Can provide an accurate measurement of CRL in the 7th to 14th weeks.
- Measurements used in the 16th to 30th weeks are biparietal diameter (BPD), abdominal circumference, and femur length.
- Can detect birth defects.
- An accurate determination of fetal size is important for managing pregnancy, especially if the mother has a small pelvis.



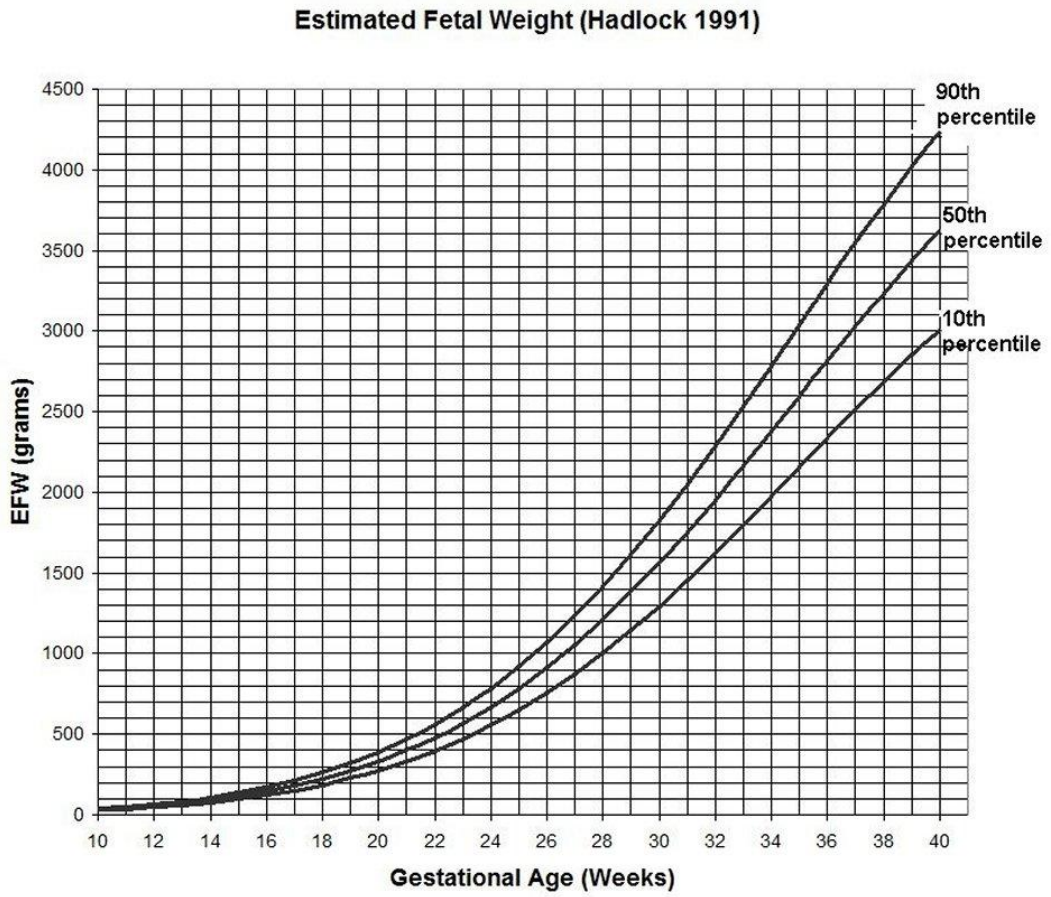




Low Birth Weight

- Low birth weight (LBW) refers to a weight <2 ,500 g, regardless of gestational age.
- Small for gestational age (SGA): the weight is less than 10th percentile for the gestational age:

  -Pathologically small (IUGR) Or -Constitutionally small (Normal)



Knowing that this is a picture of newly born twin, what is the most probable explanation of their size discrepancy?

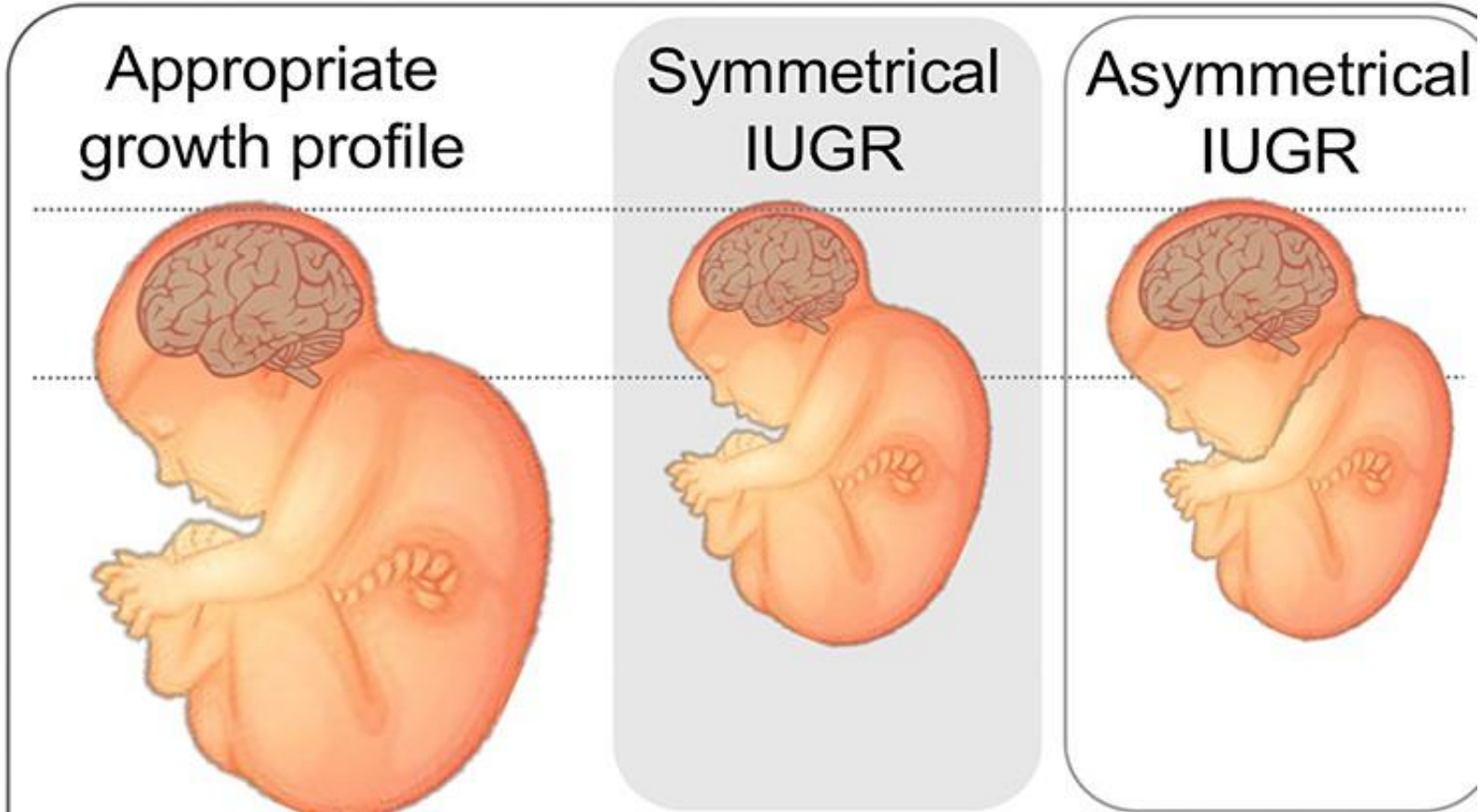
Feto-fetal transfusion



SGA AGA

IUGR (intrauterine growth restriction)

- IUGR is a term applied to infants who do not attain their optimal intrauterine growth.
- IUGR infants are pathologically small and at risk for poor outcomes.
- IUGR could be symmetrical or asymmetrical.



IUGR

Causes include:



- Chromosomal abnormalities and gene mutations.
- Teratogens.
- Congenital infections (rubella, cytomegalovirus, toxoplasmosis, and syphilis).
- Poor maternal health (hypertension and renal and cardiac disease).
- Mother's nutritional status and socioeconomic level; her use of cigarettes and alcohol.
- Placental insufficiency.
- Multiple births (e.g., twins, triplets).

IUGR

Approximately 1 in 10 babies have IUGR. They have increased risk of:

- Neurological problems.
- Congenital malformations.
- Meconium aspiration.
- Hypoglycemia.
- Hypocalcemia.
- Respiratory distress syndrome (RDS).

IUGR Long-term effects

- **What happens in the womb does not stay in the womb.**
- Babies with IUGR have been shown to have a greater chance as adults to develop **metabolic syndrome** later in life:
- Obesity.
- Hypertension.
- Hypercholesterolemia.
- Cardiovascular disease.
- Type 2 diabetes mellitus.



IUGR

- The major growth-promoting factor during fetal life is insulin-like growth factor 1 **(IGF-1).**
- **Fetal tissues express IGF-1**, and serum levels are correlated with fetal growth.
- Mutations in the **IGF-1 gene** result in IUGR, and this growth retardation is continued after birth.



Dwarfism

- **Postnatal growth** depends on growth hormone **(GH)** and growth hormone receptor **(GHR).**
- **GH deficiency** can lead to **postnatal short stature** (Dwarfism).
- This hormone binds to its receptor **(GHR)**.
- **Mutations in the GHR** result in **Laron dwarfism**, which is characterized by marked short stature, and blue sclera.
- **These individuals don't have IUGR because GH has no role during fetal development.**



Thank you

